

CSA1704 - Artificial Intelligence [SLOT A]:

1.) SMART HOME AUTOMATION:

97.5

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→ SM INTRODUCTION:

Smart home sensors are mainly used for smart home automation. Many sensors can be used for intelligent management like temperature sensors, light sensors, etc. These sensors are also used for security purposes as well.

→ SIMPLE REFLEX AGENTS:

- ✧ This type of agents doesn't use the previous records for their current decision making.
- ✧ These sensors doesn't have memory. It acts at the current changes depending on the temperature etc.

Example:

- ✧ It checks if people are there, lights are turned on.
- ✧ If not it is turned off.

These sensors takes decisions based on the current scenario and not on the past.

→ MODEL-BASED AGENTS:

This type acts based on an internal model which is made already depending on the environment. It collects information from the previous state and implements it depending on the current usage.

Eg:

- * It can track the status currently (ON/OFF).
- * The current sensors used at that time (ON/OFF).

→ ~~UTILITY-BASED~~ GOAL-BASED AGENTS:

It maintains the past records and has memory which stores the information. The actions performed will be based on a particular goal. The actions are made properly to achieve the goal. All the sensors work based on the goal.

Eg:

- * If the house is empty and locked, the sensors must automatically turn OFF everything since there is no one there.
- * If house members open the door, the appliances can be turned ON.

UTILITY-BASED AGENTS

This agent also has memory and acts based on the usage. It maintains the previous records and takes the data from it and ~~turns~~ decides whether to turn on or turn off the sensors at that time.

Eg:

- * If there ~~are~~ were 3 people in a room, fans should run in a medium speed.
- * If there are 4 people, it should automatically increase the speed.

~~CONCLUSION~~

The best way of agent usage for smart home automation is hybrid approach which combines all these A agents and runs the Smart Home Automation system.

INTRODUCTION:

When a robot is placed in an unknown maze it doesn't know the exact path or the route for its exit, neither it knows the exact direction. So, it should use searching techniques

to find the shortest path from source to destination. There are so many algorithms used to find the shortest path.

Uniform Cost Search (UCS):

This searching type acts based on the shortest path from source to destination by calculating its cost of travel.

Steps:

- It finds all possible paths or routes from the current place to reach its destination.
- It also calculates the cost covered by each of the all possible paths.
- After calculating the cost, it finally finds so many paths to reach the destination.
- It finds the shortest cost and exits finally.

Algorithm

- Starts from start nodes.
- chooses low cost path node
- Expands neighbouring node
- Repeats until the goal is reached.

G Advantages:

- * Easy to implement.
- * Lowest Path is found
- * Time saving.

Disadvantages:

- * Time consuming.
- * Error occurrence.

Iterative deepening search

It performs based on the depth and repeatedly searching the path routes.

Algorithms:

1. Search limit with 0 (depth).
2. If goal not found, increase limit to 1.
3. If goal is found, ~~exit~~ exit.

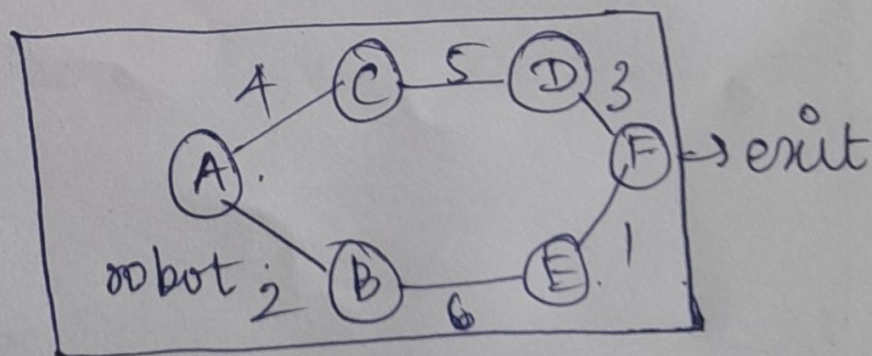
Advan:

- * less memory
- * suitable when depth is unknown.

Disadv:

- * repeats so many times.
- * less efficient.

~~All shortest path can be.~~



$$\Rightarrow A - B - E - F$$

$$\Rightarrow 2 + 6 + 1 \Rightarrow 9 //$$